

Clinical Risk Assessment Report: Patient 93

1. Primary Outcome & Risk Summary

- Target: Occult Lymph Node (OLN) Metastasis
- Model Prediction: Negative for Metastasis
- Prediction Confidence: Moderate Confidence
- Absolute Predicted Risk: 11.7% (Diagnostic Threshold: 22.1%)
- Relative Risk Multiplier: 0.53x the diagnostic threshold

Clinical Takeaway:

The machine learning model classified this patient as Low-Risk for Occult Lymph Node (OLN) Metastasis. The primary driver determining this prediction is the Small Zone Emphasis (PET) (GLZLM_SZE.PET), which reduced the patient's absolute risk by 4.3%. Biologically, this feature reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

2. Key Pathological & Imaging Drivers (Elevating Risk)

Squamous Cell Carcinoma Antigen (SCC_Ag) **Value: 0.4 [Bottom 2%] (+1.5%)**
Serum levels reflect tumor proliferation, burden, and squamous cell differentiation.

75th Percentile Tumor Density (CONVENTIONAL_HUQ3) **Value: 0.68 [Avg (68th %ile)] (+1.0%)**
Reflects the overall physical density, solid components, and general attenuation characteristics of the tumor.

3. Protective / Mitigating Factors (Reducing Risk)

Small Zone Emphasis (PET) (GLZLM_SZE.PET) **Value: -0.71 [Bottom 22%] (-4.3%)**
Reflects the proportion of small, fine texture zones, often correlating with a highly chaotic and erratic micro-architecture.

Patient Age (Age) **Value: 71.0 [Top 22%] (-1.4%)**
Patient age influences immune surveillance, tissue microenvironment, and baseline metabolic rates.

Minimum Tumor Density (CONVENTIONAL_HUmin) **Value: -0.51 [Avg (28th %ile)] (-1.4%)**
Reflects the least dense areas within the tumor on CT, often corresponding to cavitation, necrosis, or cystic degeneration.

Carcinoembryonic Antigen (CEA) **Value: 1.1 [Bottom 7%] (-0.7%)**
Elevated levels are strongly associated with increased tumor burden and potential for micrometastasis.

25th Percentile Metabolic Activity (CONVENTIONAL_SUVbwQ1) **Value: -0.31 [Avg (47th %ile)] (-0.5%)**
Reflects the baseline metabolic activity and glucose consumption of the tumor tissue.

PET Radiomic Metabolic Feature (GLZLM_ZLNU.PET) **Value: -0.48 [Bottom <1%] (-0.3%)**
Metabolic PET feature reflecting glucose avidity and cellular metabolic rate.

Machine Learning Feature Attribution (SHAP Decision Path)

